

Evaluating NBA Betting-Market Accuracy Using Game, Team, Ranking, and Player Data

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DX 699 – AI for Leaders

Exploratory Data Analysis

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1. Description of project

This project evaluates how accurately National Basketball Association betting markets describe game outcomes and whether basic team-performance information helps explain when those market expectations are correct or incorrect. The central question is not simply whether a favorite wins. Instead, the analysis distinguishes three related but different tasks: identifying the likely winner, estimating the final point margin, and estimating the combined number of points. This distinction matters because a team can win without covering the point spread, and a game can finish above or below the posted total even when the market correctly identifies the stronger team. The project therefore treats the moneyline, point spread, and total as separate forms of probabilistic or numerical forecasts.

The analysis combines four types of information. The game files contain final scores, game dates, and home and visiting team identifiers. Daily ranking files provide pregame winning percentages and home-road records. A historical betting dataset contains moneylines, spreads, totals, favorite location, and quarter-level scoring. A 2023–2024 player-statistics dataset provides additional context for possible future team-style analysis. The main game and ranking files follow the structure of Lauga's NBA games dataset, while the betting and player files came from separate Kaggle sources (Cviaxmiwnptr, 2025; Lauga, n.d.; Vivo Vinco, 2024).

The practical impact of the project is twofold. First, it demonstrates how data integration quality can determine the credibility of a sports-analytics result. Team codes, encodings, delimiters, dates, and the timing of standings information all had to be resolved before valid comparisons were possible. Second, the project provides an interpretable evaluation of market information. If favorites win frequently but point-spread and over-under outcomes remain near 50%, the market may be useful for ranking team strength while still making it difficult to obtain a systematic betting advantage. The project does not claim to prove economic market efficiency because it does not model sportsbook commission, line movement, or betting returns. It instead provides exploratory evidence about accuracy, calibration, and residual uncertainty.

2. What you learned from exploratory data analysis:

2.1 Preprocessing

The first preprocessing lesson was that the files could not be loaded as one homogeneous directory. The NBA game-data folder contained `games.csv`, `games_details.csv`, `players.csv`, `ranking.csv`, and `teams.csv`, but each file represented a different unit of observation and had a different schema. Treating the directory as one Arrow dataset caused type-inference failures because columns in one file did not correspond to columns in another. The solution was to load each table separately and document its observational level before attempting any merge.

The player files introduced a second class of formatting problems. They used semicolons rather than commas and contained Windows-encoded characters that were not valid UTF-8. Specifying a semicolon delimiter and CP1252 encoding produced separate statistical columns and preserved player names. The rankings file also contained a `RETURNTOPLAY` field with only 3,990 non-null observations out of 210,342 rows. Because the variable recorded a special

COVID-era return condition rather than routine team performance, it was removed from the cleaned rankings file.

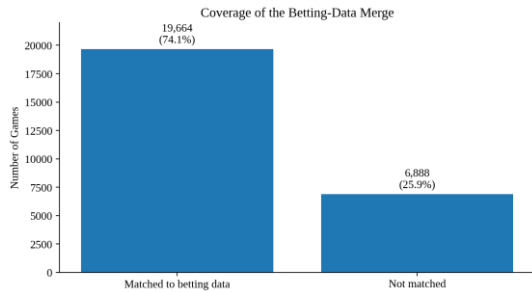


Figure 1. Betting records matched 74.06% of games.

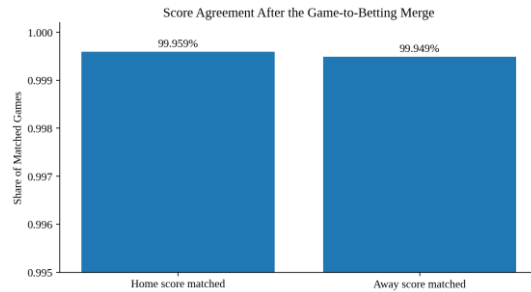


Figure 2. Score agreement exceeded 99.9% after the merge.

2.1 Preprocessing (continued)

The merge between game results and betting records used game date, standardized home-team abbreviation, and standardized away-team abbreviation. Historical team codes required explicit mappings, including sa to SAS, gs to GSW, utah to UTA, pho to PHX, and wsh to WAS. The game and team tables were first connected through numeric team identifiers; the betting table then used the standardized abbreviations. This staged approach avoided relying on team names that changed over time or appeared in multiple formats.

The resulting match rate was 74.06%. Unmatched games were not automatically classified as errors. The most likely explanation was unequal historical coverage: the betting data began in October 2007, while the game files contained a broader time range. Those unmatched games remained useful for general scoring analysis but were excluded from betting-specific comparisons. Validation used final scores only after the merge, not as merge keys. Home scores agreed in 19,656 of 19,664 matched games, and away scores agreed in 19,654. The match rates above 99.9% provide strong evidence that date and team codes connected the intended games.

Standings were joined with a backward as-of procedure so that each game received the most recent standings row available on or before the game date. This step reduced data leakage, but it does not guarantee that a same-date standings record was published before tipoff. A stricter predictive study should lag standings by one calendar day or reconstruct team records directly from prior game results. Exact duplicate rows were removed, dates were converted to date types, and records missing essential team identifiers or final scores were excluded from outcome calculations.

2.2 Univariate analysis

The univariate analysis examined each outcome without conditioning on another variable. Approximately 58.7% of games were won by the home team, which is consistent with the established NBA home-advantage literature. Ribeiro, Mukherjee, and Zeng (2016) found that NBA teams tend to score at a higher rate at home, although the size of that advantage differs among teams and has changed over time. The present result does not isolate crowd, travel, referee, or scheduling effects, but it establishes home location as an important baseline variable.

Favorites won 68.36% of validated games. That rate demonstrates that the market usually identified the stronger side, but it should not be interpreted as a 68% profitable betting rate. Moneyline prices require bettors to risk more on stronger favorites, and this analysis did not calculate profit after odds and commission. In contrast, the home team covered the point spread in 49.41% of games, and the game went over the posted total in 49.59%. These two results were extremely close to an even split.

The near-50% cover and over rates are important because sportsbooks use the spread and total to balance two-sided propositions rather than merely forecast the winner. The result is also consistent with Mateos's (2019) distinction between winning a game and covering the spread: a

winning team does not necessarily outperform the market-adjusted expectation. Pushes were treated separately when actual results equaled the posted line.

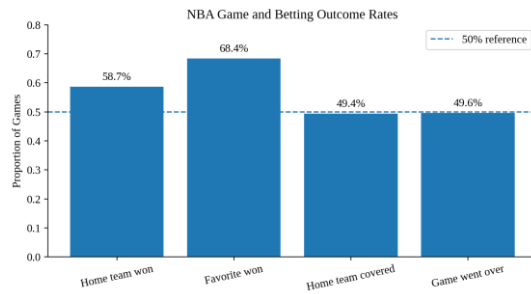


Figure 3. Favorites won often, while cover and over rates stayed near 50%.

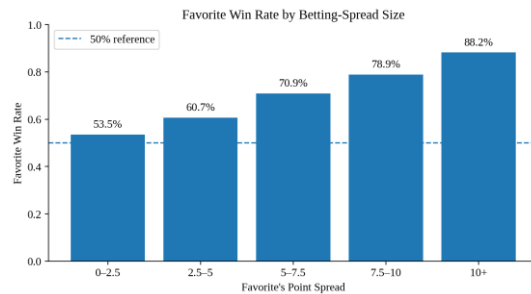


Figure 4. Favorite win rate increased with spread size.

2.2 Univariate analysis (continued)

Favorite strength produced a clear monotonic pattern. Teams favored by 0 to 2.5 points won 53.5% of games, while teams favored by 2.5 to 5 points won 60.7%. The rate increased to 70.9% for favorites of 5 to 7.5 points, 78.9% for favorites of 7.5 to 10 points, and 88.2% for favorites of at least 10 points. Thus, the spread magnitude functioned as an ordinal measure of market confidence even though the spread itself was designed to evaluate margin rather than simply winner.

The probability results should be interpreted together with uncertainty. Even strong favorites lost more than one game in ten, and small favorites were only modestly more likely to win than lose. The categories also contained different sample sizes: 3,947 games in the smallest-favorite group, 5,216 in the 2.5–5 group, 4,675 in the 5–7.5 group, 3,105 in the 7.5–10 group, and 2,706 in the 10-plus group. All groups were therefore large enough to support stable descriptive comparisons.

A key leadership implication is that a high winning percentage can coexist with poor decision value if the price already reflects the advantage. The market's ability to identify stronger

teams does not show that a user can profit by selecting every favorite. Any later model must compare its predictions with market probabilities and betting prices, rather than with a simple 50% benchmark alone.

2.3 Bivariate analysis

The bivariate analysis compared market forecasts and team-strength measures with observed game outcomes. The market-expected home margin had a Pearson correlation of 0.457 with the actual home margin. Pregame winning-percentage difference had a nearly identical correlation of 0.459 with final margin. These moderate relationships show that both the spread and a basic standings measure contained useful information, but neither explained most game-to-game variation.

The strongest relationship was between the posted betting total and actual combined points, with a correlation of 0.622. Games expected to have higher totals generally did produce more scoring. The remaining dispersion was still large, reflecting variation in pace, shooting efficiency, overtime, injuries, and game-specific randomness. The relationship between no-vig home implied probability and the binary home-win outcome was 0.407. A binary outcome naturally limits the magnitude of a simple Pearson correlation, so calibration analysis was more informative than correlation alone.

The comparison between spread and standings is particularly useful. The nearly equal correlations suggest that current winning percentage captured much of the same broad team-strength signal reflected in the betting line. However, this does not mean standings could replace the market. The spread can incorporate injuries, rest, travel, matchups, and information not represented in season winning percentage. A multivariable model should test incremental value by comparing a standings-only model, a market-only model, and a combined model.

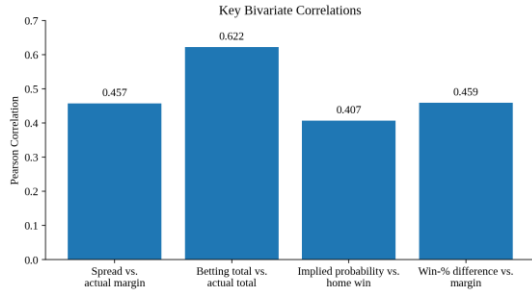


Figure 5. Betting totals had the strongest linear association with observed outcomes.

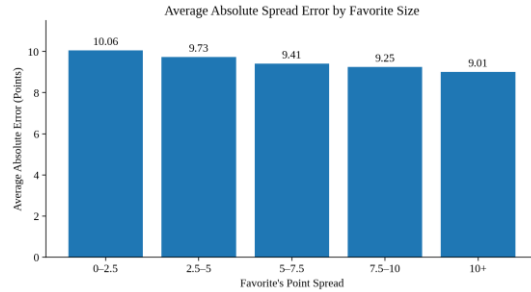


Figure 6. Exact spread errors remained near nine to ten points.

2.3 Bivariate analysis (continued)

Figure 6 compares absolute spread error across favorite categories. Even as the favorite became stronger, average error declined only from 10.06 to 9.01 points, showing that exact margins remained difficult to forecast. Figure 7 shows the distribution of actual home-team point margins. Its center slightly above zero is consistent with home advantage, while the long tails show that occasional blowouts and large game-level deviations remained part of the outcome distribution.

Moneyline calibration produced one of the most important findings. When games were grouped by the market's normalized no-vig probability for a home win, observed home-win frequencies closely followed the diagonal perfect-calibration relationship. The Brier score was 0.2023 across 19,637 games. The Brier score is the mean squared difference between a predicted probability and the observed binary outcome, so lower values indicate better probability forecasts (Brier, 1950). Calibration and Brier score should be considered together because a single score combines calibration and discrimination.

These results provide evidence that moneyline probabilities were meaningfully informative. They do not prove that the market was perfectly efficient or that no profitable subgroups existed. A formal efficiency test would require closing lines, consistent timestamps,

transaction costs, and an out-of-sample betting strategy. The present analysis is best interpreted as an evaluation of forecast quality and balance.

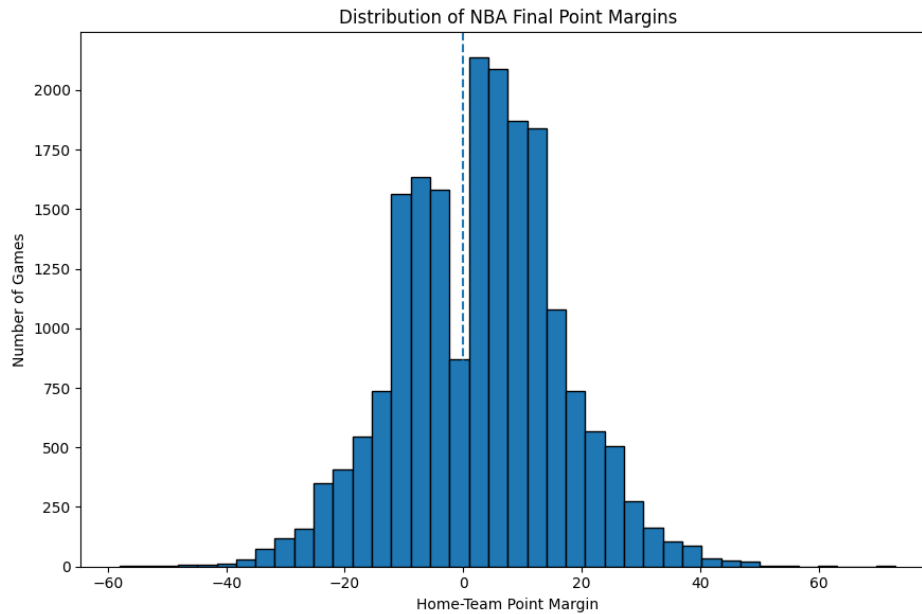


Figure 7. Final point margins were centered slightly above zero and retained long tails, showing home advantage and substantial game-level uncertainty. Source: Author's analysis.

3. Conclusions:

3.1 Expected/unexpected

Several findings were expected. Home teams won more frequently than visiting teams, consistent with prior NBA research on home advantage. Favorites also won more often as spread size increased. The smooth rise from 53.5% for small favorites to 88.2% for favorites of at least 10 points shows that market confidence and observed win probability moved in the same direction. The close alignment of moneyline probability groups with actual win rates was also consistent with the idea that betting prices aggregate public information.

The strongest unexpected result was that the betting total had a substantially stronger correlation with its observed outcome than the point spread had with final margin. The total correlation of 0.622 exceeded the spread correlation of 0.457. This suggests that league scoring

environment, expected pace, and offensive-defensive quality may be more stable or easier to aggregate than the difference between two teams' final scores. A second unexpected finding was that winning-percentage difference correlated with point margin at 0.459, almost exactly matching the spread. A simple standings measure therefore captured a surprising amount of the linear signal.

The average absolute spread error remained large in every favorite category. It declined from 10.06 points among favorites of 0 to 2.5 points to 9.01 points among favorites of 10 or more. Stronger favorites were much easier to classify as winners, but their exact margins were only modestly easier to estimate. This distinction explains why favorites won 68.36% of games while the home-cover rate remained 49.41%. The market was more successful at ranking teams than at forecasting exact realized margins.

3.2 Disqualified data

No complete dataset was disqualified, but some records and variables were restricted. The 6,888 games without a betting match remained available for general game analysis but were excluded from spread, total, and moneyline comparisons. Eight home-score and ten away-score mismatches were removed from the validated betting sample, along with three matched games missing required betting values. RETURNTOPLAY was dropped because it was mostly missing and represented an exceptional COVID-era condition. The 2023–2024 player file was retained for limited analysis but could not support the full historical study.

Final-game box-score statistics were excluded from pregame prediction because using rebounds, assists, or shooting percentages from the same completed game would create target leakage. Same-day standings also require caution; every production predictor should be verified as available before tipoff.

3.2 Suggested supervised/unsupervised analyses

Supervised analysis should begin with logistic regression for HOME_WIN, comparing market-only, team-only, and combined predictors. Gradient boosting could test nonlinear interactions. A second task should predict POINT_MARGIN with the posted spread as the baseline; linear or ridge regression would measure incremental standings value, while gradient boosting could improve flexibility. A third model could predict TOTAL_POINTS relative to the posted total. All models should use chronological splits and metrics suited to the target, including ROC-AUC, log loss, Brier score, calibration, and mean absolute error. For unsupervised analysis, standardized team-season features could be reduced with principal components and clustered with K-means to identify offensive, defensive, high-pace, and balanced profiles. Silhouette score and stability across seasons should evaluate cluster quality.

Citations

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5. AI appendix

Generative AI was used as a support tool during this project. The model used was OpenAI GPT-5.5 Thinking through ChatGPT. AI assistance debugging Python, Polars, Pandas, and PyArrow loading errors. AI was also used in formulating ideas, formatting, and grammar checking.